

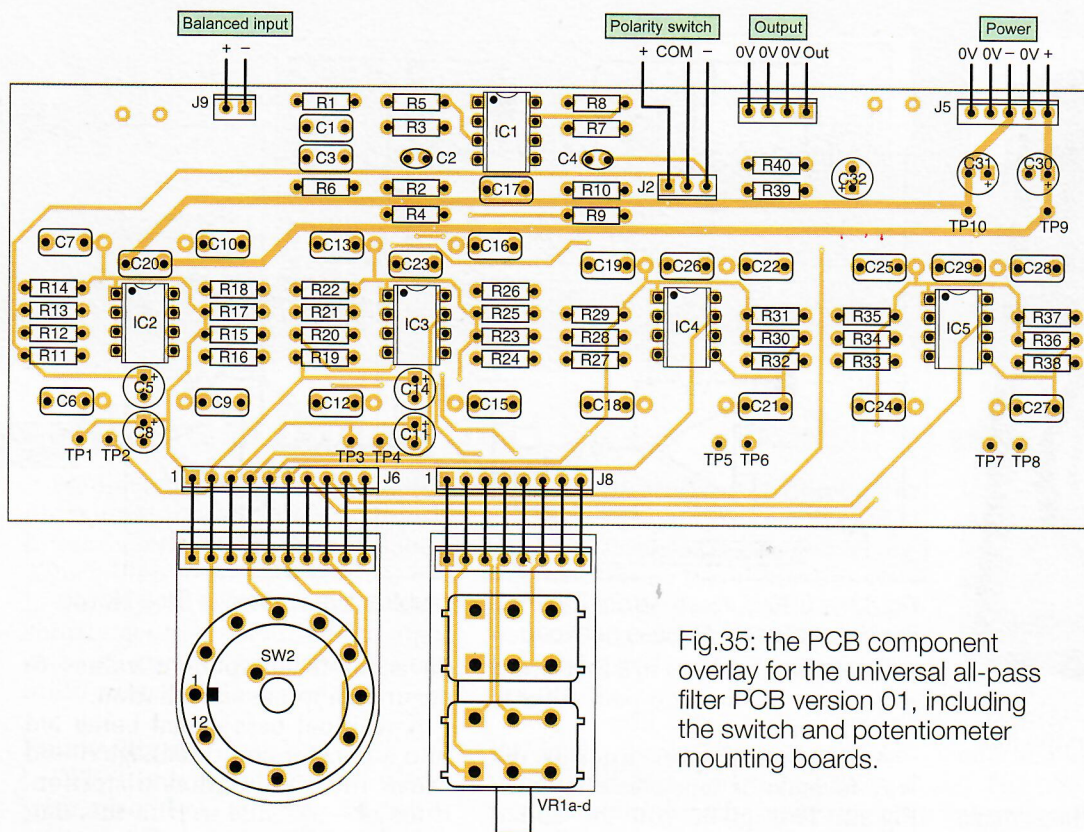


Fig.35: the PCB component overlay for the universal all-pass filter PCB version 01, including the switch and potentiometer mounting boards.

To expand on that last point, an LR crossover is designed so that each output is 6dB down at the crossover frequency. On-axis, the acoustic output of the loudspeaker at this point is the result of constructive interference between the two drivers, so the pressures simply sum.

Since the signals sent to the two drivers are identical at the crossover frequency, the two acoustic outputs add coherently, resulting in a doubling of pressure (+6dB). The -6dB level from each driver therefore sums to give a flat on-axis response. Without this -6dB level in each branch, the two drivers would produce an acoustic peak at the crossover frequency.

The off-axis behaviour is more complex. There is usually a large dip in the vertical off-axis response, depending on the driver spacing and directivity.

Soldering time

Enough theory for now; it's time to get building. I suggest using the generic op amp boards described in the December 2022, January 2023 and February 2023 issues to build the switched inverter, attenuverter and single or two stage all-pass filters. For the more complex circuits, the universal all-pass filter circuit shown in Fig.34 is needed.

This has up to eight cascaded all-pass sections or poles available, built around four dual op amps. Note that 16 stages (two units in series) are required for the effect to be clearly

heard on sounds such as kick drums, where the thump will appear to lag behind the click.

Grindle has designed a board; its overlay diagram is shown in Fig.35. There is a rotary switch to select the number of all-pass sections required (SW2) and a four-gang potentiometer (VR1) to provide tuning. It would be nice to vary all eight sections to get the biggest adjustment range possible, but I've only recently found eight-gang pots as stock items (eg, the Bourns PTD908-1015F-B103 from Mouser). Those could be used in a future design.

Both these controls can be omitted to make a fixed delay unit if desired. The provision of the pot connector (J8) allows a modulator board to be plugged in to make a sweepable studio phaser. This could use LM13700 transconductance op amps, VCAs, Vactrols or optoisolated JFETs. Ordinary JFETs won't work here, since they have to be referred to ground; these are better used for a 9V guitar phaser pedal, which we'll describe later on.

There is a balanced input section to drive the all-pass filter chain, with a polarity switch, SW1. This is built around two differential stages wired in antiparallel to give equal input impedances on both input pins.

There seemed no point in employ-

ing a dedicated single op amp (an NE5534) here. Surprisingly, the dual version is cheaper than the single version, and every audio engineer has tubes of 5532s.

Poka-yoke

The connectors on the PCB are all different sizes, in accordance with the Toyota poka-yoke assembly technique, to avoid mistakes. I blew a board up recently because the output and power supply connectors were both three-pin Molex plugs. I'm never going to make that mistake again!

The completed PCB is shown in Photo 19.

The switch and pot are on little adaptor boards shown in Photos 20 & 21. Both of these

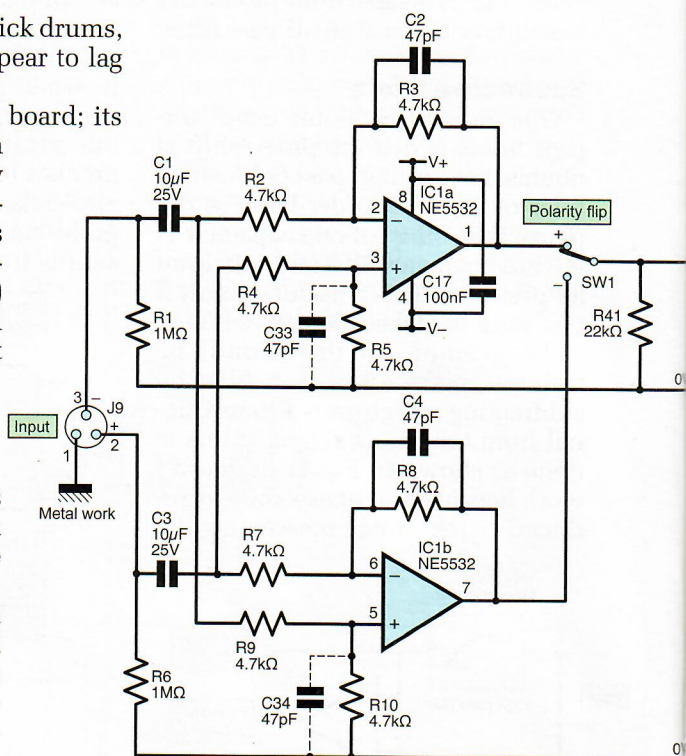
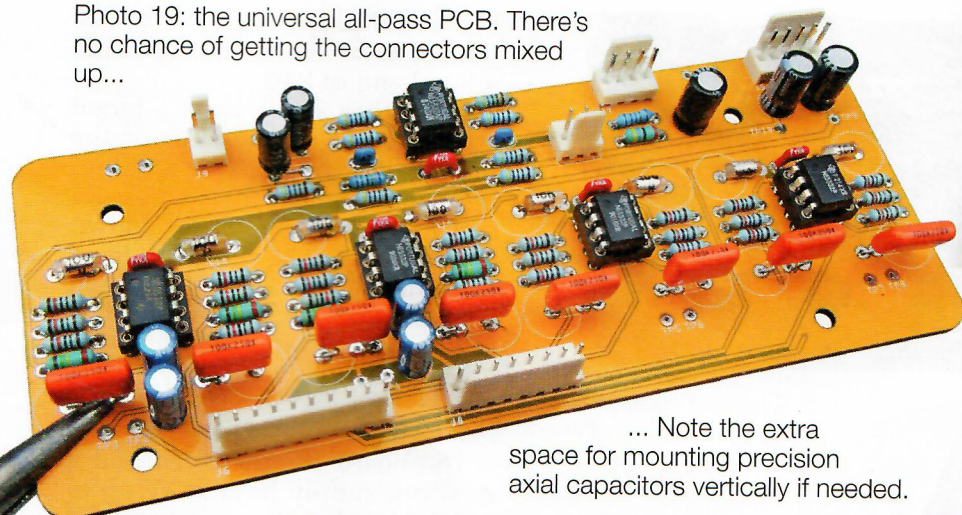


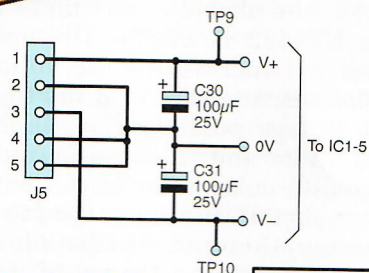
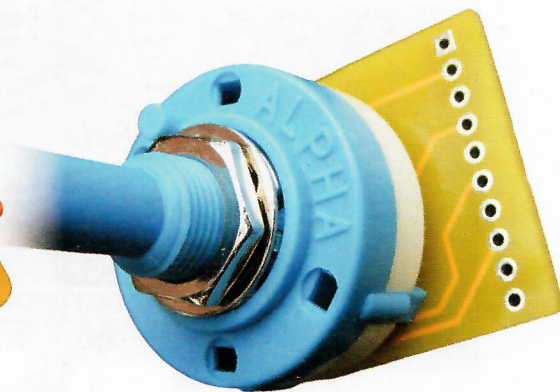
Photo 21: the quad-gang pot also has its own PCB. This could be useful for other filter projects.

Photo 19: the universal all-pass PCB. There's no chance of getting the connectors mixed up...



... Note the extra space for mounting precision axial capacitors vertically if needed.

Photo 20: the switch is mounted on a sub-board with a handy SIL header connection.



*R11, R15, R19, R23 are 150kΩ only if VR1 is fitted.

Fig.34: the universal all-pass filter circuit, enough for most of our studio needs.

